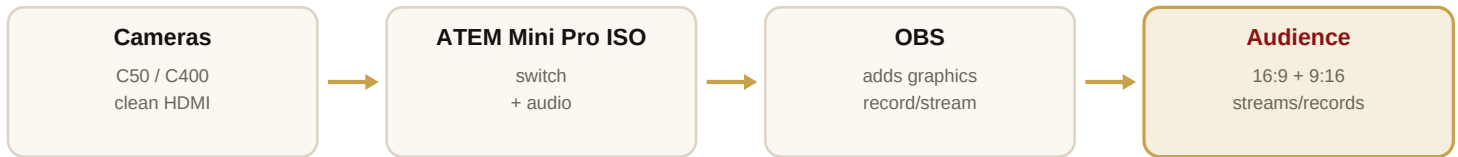


THE SIGNAL FLOW



One control drives both outputs. Graphics you push in the portal travel through a server “bus” that BOTH the 16:9 and the 9:16 OBS overlays poll, so they update together within about a second. OBS runs its own browser and can't share yours — the bus is the bridge. You never paste graphics into OBS by hand.

WHAT EACH PIECE DOES

The cameras (C50 / C400)

Send a clean 1080p picture over HDMI into the switcher.

The ATEM Mini Pro ISO

Switches cameras and mixes audio; one feed to the computer over USB-C.

OBS

Takes the ATEM feed, lays Dot 1 News graphics on top, records/streams — 16:9 and 9:16.

The editorial portal

Where a producer drives lower thirds, bug, ticker/strap, and breaking — live, for both.

CAMERAS — CANON C50 / C400

- Turn on CLEAN HDMI output so menus, focus guides, and info don't show on air (Canon HDMI output / display-off setting).
- Set every camera to the SAME resolution and frame rate: 1920x1080 at 30p (or all 60p). Never mix frame rates.
- Run each camera's HDMI into ATEM inputs 1–4. Seat connectors firmly; a locking HDMI adapter helps on a live set.
- Frame with vertical in mind: your 9:16 output is a centre crop of the 16:9 picture, so keep the subject near centre.
- Optional: let each camera also record locally as a backup while it feeds the switcher.

AUDIO

- Simplest path: plug your host mic into the ATEM's 3.5mm audio input and set its level on the ATEM.
- Or use camera audio embedded in the HDMI feed if the mic is on the camera.
- In the ATEM audio mixer, set the host mic to ON (always on). Use AFV for sources heard only when live.
- Watch levels: peaks around -12 to -6 dB. Red (0 dB) is clipping — turn it down.

ATEM MINI PRO ISO

- Connect: cameras to HDMI IN 1–4; USB-C to the computer (it appears as a webcam); optional HDMI OUT to a monitor.
- Switch: Program row buttons pick what's live. CUT is a hard cut; AUTO runs the selected transition.
- Optional ISO record: a fast USB-C SSD records every input separately for editing in Resolve later.
- The ATEM Software Control app gives deeper audio, media, and macro settings — not required for a first test.

One rule that prevents most problems:

Cameras, ATEM, and OBS should all be the same frame rate. Mixed frame rates cause stutter and sync drift.

OBS — BRING IN THE SWITCHER

- Add the ATEM feed: Sources → + → Video Capture Device → pick the Blackmagic ATEM. Set 1920x1080.
- Set the canvas: Settings → Video → Base and Output 1920x1080, FPS matching your cameras.
- Confirm audio: the ATEM audio arrives with that capture device — check it shows and moves in the Audio Mixer.

OBS — ADD THE 16:9 OVERLAY

- Sources → + → Browser Source. URL (from Broadcast → Live tools, Program URL):
- <https://editorial.dot1.media/broadcast/dot1-news-broadcast.html>
- Width 1920, Height 1080. Leave “Shutdown source when not visible” OFF.
- Drag the overlay ABOVE the ATEM source. It detects OBS and runs transparent automatically.

EDITORIAL — DRIVE THE GRAPHICS (from the portal)

- Lower thirds: Broadcast → open a rundown segment → Take to air (Clear removes it).
- Bug, ticker, breaking: Broadcast → On-air graphics → Show / Hide; edit ticker text and press Update.
- Anything you push appears on the OBS overlay within ~1 second. Producer or Owner rights required.
- Grab exact URLs any time from Broadcast → Live tools.

Same switch, same control, second output.

The 9:16 overlay reads the SAME bus, so the graphics you push drive both formats at once.

OBS — A SECOND, VERTICAL OUTPUT

- Make a vertical profile: Profile → New (e.g. “Dot 1 Vertical”) and a new Scene Collection so it stays separate.
- Set the canvas: Settings → Video → Base and Output 1080x1920 (portrait), same FPS as the 16:9 profile.
- Add the ATEM the same way (Video Capture Device), then scale/crop it to fill the tall frame — a centre cut of the 16:9 picture.
- Tip: hold Alt and drag the source edges to crop, or add a Crop/Pad filter for a clean centre 9:16.

OBS — ADD THE 9:16 OVERLAY

- Sources → + → Browser Source. URL (Broadcast → Live tools, the vertical tool's Program URL):
 - <https://editorial.dot1.media/broadcast/dot1-news-vertical.html>
- Width 1080, Height 1920. Place it ABOVE the camera source. It runs transparent in OBS automatically.

HOW THE PORTAL MAPS ONTO VERTICAL

Lower third

Portal lower third → the vertical name/role lower third (kicker is 16:9-only).

Bug

Bug on/off, colour, and LIVE dot carry over identically.

Breaking

Breaking banner carries over identically.

Ticker → Strap

The ticker drives the vertical HEADLINE STRAP (vertical has no crawl): label + first headline.

Stream the vertical profile to TikTok, Reels, or Shorts. Switch OBS profiles to move between 16:9 and 9:16; or run both at once on a machine with the horsepower using OBS multiple-output / a second instance.

Record, edit in DaVinci, then upload the finished cut.

Episodes go to Cloudflare Stream (same as live) and appear in the app and site.

THE WORKFLOW

- Record the show in OBS, or use the ATEM's ISO record to a USB-C SSD for separate camera files.
- Edit and color in DaVinci Resolve. Export a finished file (H.264/H.265 MP4 is ideal).
- In the portal: Broadcast → Publish episode. Add title, description, category, and host.
- Drop in the exported file and press Upload & publish (or Upload as draft to hold it).

WHAT HAPPENS

- The file uploads straight to Cloudflare Stream, resumable — a large export won't choke, and a hiccup resumes.
- Keep the tab open until the progress bar reaches 100%.
- Cloudflare encodes it (a few minutes) and generates the thumbnail; then it appears in the app and site.
- Manage every episode under Published → Videos (edit details or delete), same as any video.

GOOD TO KNOW

- Live and recorded share one video pipeline (Cloudflare Stream), so there's nothing extra to set up.
- Producers, editors, and owners can publish episodes.
- Export at 1080p to match the rest of the chain; very large 4K files upload slower.
- The episode plays as adaptive HLS, so it works on any connection and device.

Live streams to news.dot1.media and the app through Cloudflare Stream.

You control it from the portal: Broadcast → Go live. The recording is kept automatically.

ONE-TIME SETUP

- Maintainer: on the editorial project set CLOUDFLARE_ACCOUNT_ID, CLOUDFLARE_STREAM_TOKEN (Stream:Edit), and CLOUDFLARE_STREAM_CUSTOMER_CODE (just the code, e.g. lkaw9..., not the full hostname); redeploy.
- Producer: Broadcast → Go live → Set up live input (once). Copy the Server URL and Stream key it shows.

EACH BROADCAST

- OBS: Settings → Stream → Service “Custom”. Paste the Server into Server and the Stream key into Stream Key.
- Build your program as usual (ATEM feed + the graphics overlay), then Start Streaming in OBS.
- In the portal, press Go live and set a title. A Live surface appears on the site and app within seconds.
- Drive lower thirds, bug, ticker, breaking from On-air graphics exactly as normal — they ride in the stream.
- Press End broadcast when done. Cloudflare saves the recording, which can become a posted episode.

GOOD TO KNOW

- After OBS connects there's a 10–20 second beat before the stream is playable — refresh /live if it's black.
- Live video always runs a short delay behind real time. That's normal for live HLS, not a fault.
- The Stream key is a credential (anyone with it can push to your input). It's Producer/Owner only — keep it private.
- If the key ever leaks, ask for a new live input to be provisioned (rotates the key).

PRE-FLIGHT, IN ORDER

- ☐ Cameras: clean HDMI on, all the same 1080p frame rate, into ATEM 1–4.
- ☐ Audio: mic into the ATEM, level set, mixer channel ON, no red clipping.
- ☐ ATEM: USB-C to the computer. Cut through inputs; confirm every camera shows.
- ☐ OBS 16:9: ATEM as Video Capture Device (1080p); 16:9 overlay Browser Source on top (1920x1080); audio in the mixer.
- ☐ OBS 9:16: vertical profile at 1080x1920; ATEM cropped to centre; 9:16 overlay Browser Source on top (1080x1920).
- ☐ Portal: sign in → Broadcast → On-air graphics. Show the bug → confirm on BOTH overlays → Hide.
- ☐ Lower third: Take from a rundown segment → confirm on both → Clear.
- ☐ Ticker: Show → confirm it crawls on 16:9 and shows on the vertical strap → Hide. Breaking: Show → confirm → Hide.
- ☐ Record a 60-second test on each profile while switching cameras and pushing graphics. Review both.
- ☐ If streaming: set the stream key and go live to a private/test target first.

TROUBLESHOOTING

ATEM not listed in OBS

Re-seat USB-C; pick the ATEM in Video Capture Device; quit ATEM Software Control if it holds the device.

A camera input is black

Check HDMI seated and clean 1080p at the same fps; try another ATEM input.

Graphics don't show in OBS

Right-click the browser source → Refresh; confirm the correct Program URL (16:9 vs 9:16); confirm you pushed and have broadcast rights.

One format updates, the other doesn't

The stale one's browser source needs a Refresh, or its URL is the wrong overlay. Both read the same bus.

Overlay has a black background

Not in output mode — it must be the OBS browser source (auto), or add ?output to the URL.

Vertical looks stretched

Crop the ATEM to a centre 9:16 instead of stretching; keep the overlay source at 1080x1920.

No audio in OBS

Unmute the ATEM device in the OBS Audio Mixer; confirm the mic is ON in the ATEM mixer.